

Megha Srivastava

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My research develops **safe and reliable AI systems by addressing failures at both the computational and human levels**. I (i) build methods for verifiable and secure AI systems, and (ii) design interactive systems and algorithms that support user learning, understanding, and skill development. I enjoy working across domains, including software development, motor skill learning, and small-molecule drug discovery. My work has been recognized by an ICML Best Paper Runner-Up Award, is supported by the NSF and IBM PhD Fellowships, and appears in venues including NeurIPS, ICML, and Nature.

Education

Stanford University • Stanford, CA present
PhD in Computer Science, co-advised by Dorsa Sadigh & Dan Boneh
Supported by the NSF Graduate Research Fellowship (2018 – 2023) and IBM PhD Fellowship (2023-2024)

Massachusetts Institute of Technology • Cambridge, MA Fall 2023 – Winter 2024
Visiting PhD student at the Computer Science & Artificial Intelligence Laboratory, hosted by Jacob Andreas

Stanford University • Stanford, CA Spring 2019
BS+MS in Computer Science with Honors and Distinction, Minor in Creative Writing
Research advised by Percy Liang & Tatsunori Hashimoto
Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science
Bing Overseas Studies Program at Oxford University (Logic & Computability) in Spring 2017

Experience

Research Intern – Genesis Molecular AI Summer 2026
San Mateo, CA

- Deep learning research at the intersection of generative AI and drug discovery.
- Led to “Pyrite: A Benchmark for Protein-Ligand Pose Ranking Agents” (NeurIPS 2026 AgenticLS Workshop).

Member of Technical Staff (Contractor) – Anthropic Spring 2025- present
San Francisco, CA

- Support research and writing (e.g. Findings from a Pilot Anthropic–OpenAI Alignment Evaluation Exercise .

PhD Intern – Hudson River Trading Winter 2026
New York, NY

- Applied machine learning towards building robust financial models with the Algorithm Development team.

AI Resident – Microsoft Research Fall 2019 – Fall 2020
Redmond, WA

- Quantifying effects of nondeterminism in neural network training (Mentors: Besmira Nushi, Eric Horvitz).
- Led to “An Empirical Analysis of Backward Compatibility in Machine Learning Systems” (KDD 2020).

Research Intern – Google Research Summer 2019
Los Angeles, CA

- Improving robustness of image understanding models with the Research & Machine Intelligence team.

Research Intern – ETH Zürich Learning & Adaptive Systems Group Summer 2018
Zürich, Switzerland

- Active learning and adaptive hypothesis testing (Mentors: Hoda Heidari, Andreas Krause)
- Supported by the ETH Zürich Student Summer Research Fellowship.
- Led to “Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning” (KDD 2019).

Research Assistant – Vision & Perception Neuroscience Group Summer 2016
Stanford, CA

- Generalization and perceptual invariances in both human and artificial vision models (Mentor: Kalanit Grill-Spector)
- Supported by the Bio-X Undergraduate Research Award.

Preprints

Megha Srivastava, Michael H. Lim, Allison M. Keys, Nate Gruver, Pranav Murugan. “Pyrite: A Benchmark for Protein-Ligand Pose Ranking Agents”, NeurIPS Agentic AI for Biological Discovery Workshop, 2026
[AI for Science](#)•[Robust ML](#)

Megha Srivastava, Jonathan Ouyang, Eric Zhou, Andrew Silva, Emily Sumner, Dorsa Sadigh, Yuchen Cui, Deepak Gopinath, Guy Rosman. “Proximal State Nudging: Reducing Skill Atrophy from AI Assistance”, ICML Trustworthy AI for Good Workshop, 2026
[AI for Education](#)•[Embodied AI](#)

Erez Badash $\alpha\beta$, Dan Boneh $\alpha\beta$, Ilan Komargodski $\alpha\beta$, **Megha Srivastava** $\alpha\beta$. "Hawkeye: Reproducing GPU-Level Non-Determinism", Ninth Annual Conference on Machine Learning and Systems (MLSys), 2026
[Robust ML](#)·[Security](#)

Alexis Ross*, **Megha Srivastava***, Jeremiah Blanchard, Jacob Andreas. "Modeling Student Learning with 3.8 Million Program Traces", International Conference on AI in Education (AIED), 2026
[AI for Education](#)·[NLP](#)

Jan Betley*, Niels Warncke*, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martín Soto, **Megha Srivastava**, Nathan Labenz, & Owain Evans "Training Large Language Models on Narrow Tasks Can Lead to Broad Misalignment"
Nature, 2026
[Robust ML](#)·[NLP](#)

Megha Srivastava, Cédric Colas, Dorsa Sadigh, Jacob Andreas. "Policy Learning with a Language Bottleneck"
Transactions in Machine Learning (TMLR), 2026
[NLP](#)·[Embodied AI](#)·[Human-AI Interaction](#)

Megha Srivastava*, Reihaneh Iranmanesh*, Yuchen Cui, Deepak Gopinath, Emily Sumner, Andrew Silva, Laporsha Dees, Guy Rosman, Dorsa Sadigh. "Shared Autonomy for Proximal Teaching", ACM/IEEE International Conference on Human-Robot Interaction (HRI), 2025.
[AI for Education](#)·[Embodied AI](#)

Megha Srivastava, Simran Arora, Dan Boneh. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism", Advances in Neural Information Processing Systems 38 (NeurIPS) , 2024.
[Robust ML](#)·[Security](#)

Neil Perry*, **Megha Srivastava***, Deepak Kumar, Dan Boneh. "Do Users Write More Insecure Code with AI Assistants?" ACM Conference on Computer and Communications Security (CCS), 2023.
[mlsec.org Top-100 Computer Security Papers](#) 
[Security](#)·[Human-AI Interaction](#)

Mina Lee, **Megha Srivastava**, Amelia Hardy, John Thickstun, Esin Durmus, Ashwin Paranjape, Ines Gerard-Ursin, Xiang Lisa Li, Faisal Ladhak, Frieda Rong, Rose E. Wang, Minae Kwon, Joon Sung Park, Hancheng Cao, Tony Lee, Rishi Bommasani, Michael Bernstein, Percy Liang "Evaluating Human-Language Model Interaction" Transactions of Machine Learning Research (TMLR), 2023.

Led project experimental design; follow-up analysis and dataset release with John Thickstun ()

§ **Question Answering**: Megha Srivastava, John Thickstun, Rose Wang, Minae Kwon, Mina Lee

§ **Crossword Puzzles**: Megha Srivastava

[Human-AI Interaction](#)·[NLP](#)

Megha Srivastava, Noah Goodman, Dorsa Sadigh. "Generating Language Corrections for Teaching Physical Control Tasks" Proceedings of the 40th International Conference of Machine Learning (ICML) , 2023.
[AI for Education](#)·[Embodied AI](#)

Megha Srivastava, Erdem Biyik, Suvir Mirchandani, Noah Goodman, Dorsa Sadigh. "Assistive Teaching of Motor Control Tasks to Humans" Advances in Neural Information Processing Systems 36 (NeurIPS) , 2022.
[AI for Education](#)·[Embodied AI](#)

Siddharth Karamcheti*, **Megha Srivastava*** Percy Liang, Dorsa Sadigh. "LILA: Language-Informed Latent Actions" Conference on Robot Learning (CoRL), 2021.
[Embodied AI](#)·[NLP](#)

Megha Srivastava and Noah Goodman. "Question Generation for Adaptive Education" Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics (ACL), 2021.
[AI for Education](#)·[NLP](#)

Megha Srivastava, Tatsunori Hashimoto, Percy Liang. "Robustness to Spurious Correlations via Human Annotations" Proceedings of the 37th International Conference on Machine Learning (ICML), 2020.
[Robust ML](#)·[NLP](#)·[Human-AI Interaction](#)

Megha Srivastava, Besmira Nushi, Ece Kamar, Shital Shah, Eric Horvitz "An Empirical Analysis of Backward Compatibility in Machine Learning Systems" Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2020.
[Robust ML](#)

Publications Cont. (See Google Scholar for full list, * denotes equal contribution)

Megha Srivastava, Hoda Heidari, Andreas Krause. “Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning.” Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 2019.

Human-AI Interaction

Tatsunori Hashimoto, **Megha Srivastava**, Hongseok Namkoong, Percy Liang. “Fairness Without Demographics in Repeated Loss Minimization.” Proceedings of the 35th International Conference on Machine Learning (ICML), 2018.

Best Paper Runner-Up Award

Robust-ML

Other Works (Blog Posts, Abstracts, Technical Reports, etc.)

- **Megha Srivastava** and Claude Sonnet 3.5. “Echo: A multi-agent AI system for patient-centered pharmacovigilance.” 1st Open Conference of AI Agents for Science, 2025. **Spotlight Talk**. [Experimental conference exploring human-AI collaboration in research. More details in blog post on research process [🔗](#)]
- Sam Bowman, **Megha Srivastava**, Jon Kutasov, Rowan Wang, Trenton Bricken, Benjamin Wright, Ethan Perez, and Nicholas Carlini. “Findings from a Pilot Anthropic–OpenAI Alignment Evaluation Exercise” Anthropic, 2025. [🔗](#)
- Rose Wang and **Megha Srivastava**. “Productive Struggle: The Future of Human Learning in the Age of AI.” The Stanford AI Lab Blog, 2025. Featured and translated in Spanish by the Inter-American Development Bank. [🔗](#)
- **Megha Srivastava** and John Thickstun. “Observations from HALIE: A Closer Look at Human-LM Interactions in Information-Seeking Contexts.” Center for Research on Foundation Models Blog, 2023. [🔗](#)
- Kristin Lauter ^{$\alpha\beta$} , Cathy Yuanchen Li ^{$\alpha\beta$} , Krystal Maughan ^{$\alpha\beta$} , Rachel Newton ^{$\alpha\beta$} , **Megha Srivastava** ^{$\alpha\beta$} . “Machine learning for modular multiplication.” Proceedings of the WIN6 Workshop at Banff International Research Station for Mathematical Innovation and Discovery, 2023.
- **Megha Srivastava**, David Remus, Kalanit Grill-Spector. “The Role of Learning in Complex Object Recognition and Discrimination Across Spatial Transformations: An Experimental Comparison of Artificial CNNs and Human Subjects.” Vision Sciences Society (VSS), 2017.

Selected Awards & Fellowships

- Human Robot Interaction (HRI) Pioneers, 2025
- Rising Star in Machine Learning (University of Maryland), 2023
- IBM PhD Fellowship, 2023
- American Association for the Advancement of Science (AAAS) Mass Media Fellowship (declined), 2019
- International Conference on Machine Learning (ICML) Best Paper Runner-Up Award, 2018
- National Science Foundation Graduate Research Fellowship, 2018
- Ben Wegbreit Prize for Best Undergraduate Honors Thesis in Computer Science (Stanford), 2018
- Phi Beta Kappa, 2018
- Tau Beta Pi, 2018
- Bio-X Research Award (Stanford), 2016
- Lunsford Award for Oral Presentation of Research Finalist (Stanford), 2016

Service

- Science Small Groups Mentor (2024)
- Stanford AI Lab Blog Editor, 2020-present
- Student Program Chair for Women in Machine Learning (WiML) 2023, co-located with NeurIPS 2023
- Reviewer for NeurIPS (2020 - curr.), ICLR (2021 - curr.), ICML (2021 - curr.), RA-L 2022, CoRL 2023, L4DC 2024, CHI 2025, HRI-LBR 2026

Teaching and Mentoring

- Course Assistant for CS 224R (Deep Reinforcement Learning, 2026)
- Course Assistant for CS 255 (Introduction to Cryptography, 2025)
- Course Assistant for CS 221 (Introduction to Artificial Intelligence, 2018)
- Instructor for Stanford Splash (Demystifying Hot Topics in Computer Science, 2017)
- Mentored research of undergraduate and MS students: *Reihaneh Iranmanesh* (Amherst College → PhD at Georgetown University, co-author on HRI’25 paper), *Zhiyin Lin* (Stanford → OpenAI, CURIS program), *Mallika Parulekar* (Stanford), *Jonathan Ouyang* (UCLA), *Benita Wong* (Stanford)

Guest Lectures

1. "Security of Modern Machine Learning Systems" *AI4All*, June 2025
2. Beyond Instruction Following: Language and Human-Robot Interaction *Stanford University* (Guest Lecture for CS 329X: Human-Centered NLP), October 2024
3. "Do Users Write More Insecure Code with AI Assistants?" *University of Maryland College Park* (Guest Lecture for Large Language Models, Security, and Privacy seminar), October 2023

Invited Talks

1. "Modeling Student Learning with 3.8 Million Program Traces" *Together AI*, February 2026
2. "Modeling Student Learning with 3.8 Million Program Traces" *Deep Learning: Classics and Trends (ML Collective)*, November 2025
3. "Security of Modern Machine Learning" *Accenture Responsible AI Course*, September 2025
4. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Google Privacy in ML Seminar*, February 2025
5. "Security of Modern Machine Learning" *DaVita x Stanford Human-Centered AI*, February 2025
6. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *AI and Cryptography Workshop, Joint Mathematics Meeting*, January 2025
7. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Washington Security Seminar*, January 2025
8. "New Challenges of Trust with Large-Scale AI Systems" *University of Chicago*, November 2024.
9. "Security of Modern Machine Learning" *Stanford Cybersecurity & Privacy Festival*, October 2024
10. "Policy Learning with a Language Bottleneck" **Spotlight Talk** at *Training Agents with Foundation Models Workshop (RLC 2024)*, August 2024.
11. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *University of Toronto*, July 2024
12. "Security of Modern Machine Learning" *Accenture Responsible AI Course*, July 2024
13. "Optimistic Verifiable Training by Controlling Hardware Nondeterminism" *Stanford Security Forum*, April 2024
14. "Challenges in Human-AI Interaction for Information-Seeking Tasks" *Rising Stars in Machine Learning Workshop*, November 2023
15. "Robustness to Spurious Correlations via Human Annotations" *Two Sigma PhD Symposium*, June 2023.
16. "Do Users Write More Insecure Code with AI Assistants?" *UC Berkeley Security Seminar*, May 2023
17. "Assistive Teaching of Motor Control Tasks to Humans" *Stanford Collaborative Haptics in Robotics and Medicine Lab*, May 2023.
18. "Assistive Teaching of Motor Control Tasks to Humans" *Simons Institute Workshop on AI & Humanity*, July 2022.
19. "Assistive Teaching of Motor Control Tasks to Humans" *SystemX Alliance Fall Conference*, November 2022.
20. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Oxford University Algorithms at Work Reading Group*, May 2021.
21. "Mathematical Notions vs. Human Perception of Fairness: A Descriptive Approach to Fairness for Machine Learning" *Microsoft Research AI & Society Reading Group*, October 2019.
22. "Fairness & Robustness with Missing Information" *Stanford Causality & Cognition Lab*, December 2020.